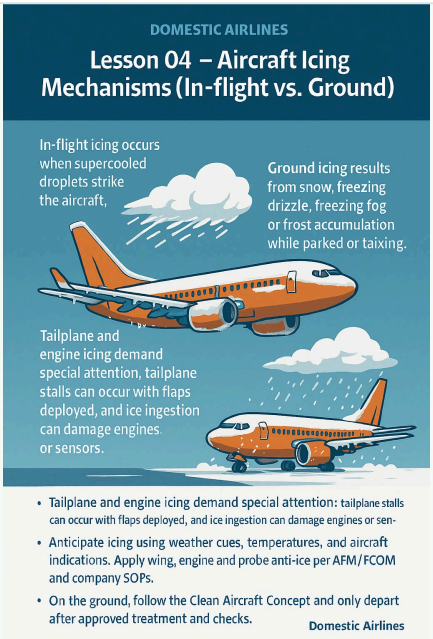
MODULE 1 · FOUNDATIONS

Lesson 04: Aircraft Icing — On the Ground and in Flight

Lesson 4 of 30 · about 8 min

LEARNING OBJECTIVES

- Distinguish ground icing from in-flight icing.
- State the standard definition of icing conditions used for engine and wing anti-ice.
- Describe the main types and intensities of in-flight icing.



Ground icing and in-flight icing have different sources and different defences.

Ground icing

On the ground the aircraft is exposed to precipitation, frost, freezing fog and slush from taxiways for as long as it is parked or taxiing. Ground icing is managed with **de-icing and anti-icing fluids** and inspections. The aircraft's own ice protection systems are **not designed** to clean wings on the ground.

Main sources:

- Frost and freezing fog on parked aircraft;
- Falling snow, freezing rain, drizzle and ice pellets;
- Cold-soaked fuel frost and clear ice;
- Slush and spray from taxiways and runways;
- Blowing snow and snow blown by other aircraft's jet blast.

Icing conditions — the standard definition

> **DEFINITION (TYPICAL AFM WORDING)**
>
> Icing conditions exist when the outside air temperature (on the ground and for take-off) or total air temperature (in flight) is **10 °C or below**, and visible moisture in any form is present — cloud, fog with visibility of 1 600 m (1 statute mile) or less, rain, snow, sleet or ice crystals.
>
> On the ground, icing conditions also exist when the OAT is 10 °C or below and there is snow, ice, standing water or slush on ramps, taxiways or runways that may be ingested by the engines or freeze on engines, nacelles or engine sensor probes.

This definition tells you when to use **engine anti-ice** (and, in flight, wing anti-ice according to the AFM). Always use the exact wording and limits in your own AFM/FCOM.

In-flight icing

Type	Conditions	Appearance
Rime	Small droplets, colder temperatures, stratiform cloud	Opaque, white, rough; forms on leading edges
Clear (glaze)	Large droplets, temperatures near 0 °C to about −10 °C, cumuliform cloud or freezing rain	Transparent, dense; can run back and form horns or ridges
Mixed	Combination of the above	Rough and partly clear
Supercooled large droplets (SLD)	Freezing drizzle and freezing rain	Ice forms **behind** the protected areas, where ice protection systems cannot remove it

Pilots report icing intensity as **trace, light, moderate or severe**. Severe icing is a rate at which the ice protection systems cannot reduce or control the hazard; immediate exit from the conditions is required.

Why the distinction matters

Ground contamination affects the whole upper surface of the wing and tail, which is where lift is generated. In-flight ice protection systems only protect leading edges, engine inlets, probes and windows. A wing that leaves the ground contaminated cannot be cleaned by switching on anti-ice.

KEY TAKEAWAYS

- Icing conditions: OAT/TAT 10 °C or below with visible moisture, or contaminated surfaces on the ground.
- Ground icing is removed with fluids — not with the aircraft's ice protection systems.
- SLD forms ice behind protected areas; exit the conditions.

KNOWLEDGE CHECK

On the ground the OAT is +8 °C and there is slush on the taxiways but no precipitation. Are icing conditions present?

A. No — there is no visible moisture in the air

B. Yes — slush that may be ingested at OAT 10 °C or below counts as icing conditions

C. Only if the temperature falls below 0 °C

D. Only at night

Answer: B. On the ground, snow, ice, standing water or slush on surfaces that may be ingested, with OAT 10 °C or below, meets the typical icing conditions definition.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.